

CrackID: A Within-Visit Benchmark for Crack Re-Identification, and the Limits of Appearance Matching

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Abstract—Maintenance systems turn photographs into tracked defect records: a re-photographed crack must match its record or be filed as new. Crack *detection* is mature, but we know of no benchmark for crack *re-identification*. We introduce CRACKID: 140 photographs of 17 walls with 212 crack identities, each wall captured in one continuous walk-around, so the benchmark measures within-visit viewpoint and scale robustness, not cross-visit re-identification. Every metric is reported against its chance level. We benchmark ten zero-shot crop matchers from four families. Ranking separates them (Rank-1 0.370–0.613 on the $n_Q=305$ rerun; prior chance 0.336 TBD), but on the thresholded same-crack decision none exceeds the prior 0.305 pairwise- F_1 floor by more than 0.069 (new F_1^* 0.337–0.374; floors TBD). Adding wall context raises OSNet’s open-set rate from 0.168 to 0.404 (prior snapshot), yet inpainting the crack out retains 0.393; context retrieves on location, not crack appearance. A geometric reference that registers on wall texture with cracks masked and Chamfer-matches projected masks reaches DIR@FAR=0.1 of 0.768 against 0.384 for the best new crop method (SuperGlue) and 0.000 for a coverage-only control, answering 48% of pairs (geometry/coverage prior snapshot, not numerically comparable to the rerun). It fails on two smooth-plaster walls and shares homography-based provenance with the annotations, so the advantage is provenance-conditioned. On a small single-seed synthetic revisit check, OSNet+context outperforms geometry in ranking and open-set rate. All rows are point estimates on 19 test identities; we claim the pattern, not the digits.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Automated building-maintenance systems convert photographs into tracked defect records: a re-photographed defect must eventually be *re-identified*, matched to its record, or given a new one. This paper first studies within-visit re-identification under viewpoint change. Wall cracks are a common and difficult case, because the same crack is re-photographed from different distances, angles, and lighting, and grows between visits.

Crack analysis is a mature *detection* and *segmentation* task, and its datasets, CrackSeg9k [1] and OmniCrack30k [2], carry no cross-observation identity labels, because the task they serve needs none. We know of no benchmark for crack *re-identification*; the natural approach, crop-and-match as in person and vehicle re-ID [3], [4], assumes a crack has stable, locally distinctive appearance, and whether it does has never

been measured. We introduce CRACKID, 140 photographs, 17 walls, 212 crack identities, to make that re-identification measurement (Fig. 2). Capture is one continuous walk-around per wall, so what is measured is *within-visit* viewpoint and scale correspondence, a hypothesized upper bound on multi-visit performance, not a measurement of it; we add a synthetic revisit protocol as a controlled secondary check (Sec. VII-D). That check uses a distinct HYBRID-SKEL pipeline: it extends the Table II REG+CHAMFER pure registration-plus-Chamfer baseline with a skeleton-comparison fallback and per-pair blending ($\tau=8.0$ px), so its reversal does not directly contradict Table II.

Appearance buys almost nothing on the decision. On ranking the ten zero-shot matchers separate (Rank-1 0.370 to 0.613 on the new $n_Q=305$ rerun; prior chance 0.336 TBD); on pairwise F_1 , the *same-crack-or-not* decision a record makes at a threshold, new rows land between +0.032 and +0.069 above the prior 0.305 best-constant floor (F_1^* 0.337–0.374; floors TBD), and at a validation-calibrated threshold, the nine new rows cluster at 0.331–0.338. With no wall-level intervals, no fine ranking is claimed.

Two controls say where the difficulty is not (prior $n_Q=280$ snapshot, pending rerun). Giving an embedder the surrounding wall raises OSNet’s open-set rate from 0.168 to 0.404; *erasing the crack* from that view costs almost none of it (0.393). Such a method retrieves on the wall, and the cue that works, even implicitly, is *location*.

A geometric baseline makes location explicit (prior snapshot, not numerically comparable to the rerun). Registering on non-crack texture, projecting crack masks into a common frame, and Chamfer-matching under global assignment reaches DIR@FAR=0.1 of 0.768 against 0.384 (best new crop, SuperGlue) and 0.404 (best prior context), answering 48% of pairs; a COVERAGE-ONLY control scores 0.000 on the same metric, so the open-set gap is matching, not alignment coverage. This is a conditional, provenance-limited result, not a deployment-ready system: two test walls have zero registration success, and the annotation and baseline both use homography-based alignment. The resulting geometric advantage is therefore a hypothesis about this benchmark, not an independently established identity effect.

The revisit check reverses the ranking story. The real data

are one continuous walk-around per wall, minutes apart, not separate visits. Under the small-warp synthetic protocol (affine $1.08\times$, 6° , 4° ; one seed; five walls), OSNet+context beats the HYBRID-SKEL at R@1 (0.826/0.671 versus 0.275/0.274) and open-set rate (0.739/0.663 versus 0.343/0.342) for growth/appearance change; the HYBRID-SKEL’s small assignment- F_1 advantage is not evidence that geometry solves real multi-visit re-identification.

Contributions: (i) CRACKID with an explicit annotation-provenance report, 45% of identities formed by an automatic merge, reported rather than assumed (Sec. III-B); (ii) a ten-method zero-shot benchmark against stated chance levels, with crop+context and crack-erased controls (Sec. VI); (iii) a small single-seed synthetic revisit checkpoint isolating appearance change from damage evolution, the only multi-visit evidence this paper has (Sec. VII-D); (iv) a geometric reference baseline (REG+CHAMFER) with coverage-aware metrics, unanswerable queries, and an admission gate, kept distinct from the synthetic HYBRID-SKEL variant (Sec. V, IV, VII).

II. RELATED WORK

Crack analysis is overwhelmingly focused on detection and segmentation [5]–[7]; its datasets annotate cracks within single images and the large consolidations merging [1], [2] inherit that scope, and multi-view crack pipelines fuse several photographs of one inspection into a surface or a segmentation [8] without solving instance correspondence between them. Structural-health monitoring does compare cracks across time with machinery close to ours [9]–[12], but identity is always *given by construction* (fixed camera, marked location, operator at a viewpoint): the subject is one known crack, the question of width or extent, and no dataset supports method comparison on the association decision itself. Those systems therefore solve measurement after correspondence has already been supplied; they do not decide whether a newly observed crack is the same physical defect as a gallery crack under an open-set protocol. The benchmark targets that missing association decision, while its real capture scope remains within one walk-around rather than a true longitudinal study.

Object re-identification [4] with open-set identification [13] is the formal setting we adopt; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and region embeddings [3], [18]–[21] apply directly to crack crops, and our benchmark measures all of them zero-shot, without CrackID fine-tuning. We cite LightGlue [22] and place recognition (NetVLAD [23]) as the natural retrieval front end, but benchmark neither given compute and scope limits and the within-visit focus; they remain future baselines, not claimed comparisons. The YOLOv8 row uses the `yolov8n.pt` detector backbone (layer 9) as an embedding [24], version-pinned in code, not a fine-tuned crack model. Vehicle-damage assessment already re-identifies damage across claims in curated settings [25], a task the open-set protocol here mirrors; the rest of that literature treats damage detection and severity rating rather than association [26]–[28]. Whether the crop scope is

itself the limiting choice for cracks is the question the benchmark is built to answer. The move the baseline makes, aligning two captures and then reasoning about what is present in one and absent in the other, is change detection [29], [30]; planar registration is classical [31]–[33] with photometric alignment (ECC [34]) as the fallback. The contribution here is not a new estimator but the observation that the alignment target and the matching target should be *disjoint*: the homography is driven by the wall precisely because the crack is what is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The benchmarked snapshot contains 140 photographs of 17 exterior walls acquired at one site over four half-days with two consumer handsets, at 3072×4080 . The walls span two surface families that behave very differently under keypoint matching: stippled render (densely textured) and smooth painted plaster, where the crack is frequently the only structure in frame. Walls split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities). Statistical power is governed by identities visible in two or more photographs, since only these form a query with an answer: 18 in validation, 19 in test. Each wall was captured in one continuous walk-around (consecutive frames a median of 3 s apart), so the dataset measures viewpoint and scale correspondence within one visit, a hypothesized upper bound on multi-visit performance, not yet a measurement of it.

Adjacent-frame retrieval. For 100% of the 280 answerable test queries in the prior snapshot, the nearest correct gallery entry is the adjacent frame (gap-sweep recount TBD for $n_Q=305$). Ranking therefore rewards short-baseline correspondence; Sec. VII sweeps the exclusion gap (answerable queries $280\rightarrow 205\rightarrow 149\rightarrow 105$) and reports that Rank-1 stays flat while the pool and prevalence change with it, so the sweep characterises robustness, not proof that duplicates are irrelevant.

B. Annotation, and why it needs auditing

One click point per crack is propagated across consecutive photographs through an ORB homography and one-to-one assignment; identities whose components appear to be fragments of one physical crack are then merged. The merge is where risk concentrates: a wrongly joined merge converts a hard negative into a free positive and inflates *every* method’s mAP together, so it cannot be caught by comparing methods. We therefore report its extent rather than assume its correctness: 95/212 identities (45%) were formed by a merge (48% on test), bridging a median maximum gap of 228 px; the adjusted Rand index against the pre-merge labeling is 0.080, meaning the merge restructured the partition rather than refining it. The triage is complete: every one of the 1,023 clicks no component claimed was reviewed against the imagery — 916 sit on visible cracks the segmenter missed rather than labelling mistakes, leaving 107 clicks (10.5%, 95% CI [8.7%, 12.5%]) naming no crack (dataset/labels/_unresolved.json). The 2,095

clicks resolve to 1,051 connected components; “points” below means clicks, not components. Single-annotator; agreement unmeasured, so the merge risk above is bounded by reporting, not removed. Counts in this section and Table I refer to the frozen benchmark snapshot (legacy directory `banchmark_out/`, historic spelling retained for reproducibility); `dataset/labels/` has since been re-edited (walls 18–19 added, audit in progress) and its raw identity/point totals no longer match exactly. The snapshot, its `result.txt`, `chance.json`, `pair_outcomes.json`, and `viewpoint.json` are the version of record for all accuracy rows.

IV. BENCHMARK PROTOCOL

The benchmark evaluates open-set retrieval: a photographed crack may or may not have a corresponding gallery instance. The protocol retains unknown cracks, calibrates thresholds on validation walls, and charges methods for unscorable pairs.

A *wall* is one physical wall at the capture site; all gallery candidates for a query come from that wall. An *identity* is a dataset label for a crack tracked across photographs, not an independently audited physical-identity judgment. A *query* is one annotated crack instance in a photograph. Its *gallery* contains instances from eligible photographs of the same wall, excluding the query photograph and capture session. A query is *known* if at least one valid gallery instance has the same identity; otherwise it is *unknown*. Test contains 280 known and 258 unknown queries in the prior snapshot (new rerun: $n_Q=305$ answerable queries; full recount of positives/valid pairs, prevalence, and floors TBD pending `chance_baseline.py` rerun). A *query-gallery pair* is one query compared with one eligible gallery instance; an *image pair* is two photographs of the same wall and is the unit charged for registration. Table I summarizes the pair and image-pair counts. On test, registration is attempted for all 301 same-wall image pairs; 255 contain at least one scored crack instance and produce validity-mask cells, while 46 yield no valid query-gallery pair after the 200px size filter (see Sec. VII-B).

A. Metrics

Ranking uses Rank-1, Rank-5, and mAP over known queries. Pairwise F_1 classifies each valid query-gallery pair. Assignment F_1 (F_1^a in Table III) evaluates the one-to-one matches made independently for each image pair. We distinguish F_1^* (oracle threshold on test, a ceiling no deployment can pick), $F_1^{@v}$ (threshold calibrated on held-out validation walls, the deployable number), and F_1^a (assignment at the validation threshold). At threshold t , $\text{FAR}(t)$ is the fraction of unknown queries whose highest finite gallery score is at least t ; $\text{DIR}(t)$ is the fraction of known queries whose highest-scoring gallery instance is both correct and at least t . The reported $\text{DIR@FAR}=0.1$ is the largest DIR among thresholds whose FAR is at most 0.1 (a conservative open-set operating point; unknown queries with no finite score count as correct rejections, known queries with no finite score count as misses).

TABLE I
COUNTING UNITS FOR EVERY METRIC (PRIOR $n_Q=280$ SNAPSHOT; FULL RECOUNT TBD FOR THE $n_Q=305$ RERUN). PER SPLIT: WALLS, PHOTOGRAPHS AND IDENTITIES; IDENTITIES IN TWO OR MORE PHOTOGRAPHS (THE ONLY ONES THAT CAN FORM A QUERY WITH AN ANSWER); ANSWERABLE QUERIES (RANKING POOL); POSITIVE AND VALID QUERY-GALLERY PAIRS (PAIRWISE- F_1 POOL, TEST PREVALENCE 0.180 PRIOR); AND SAME-WALL IMAGE PAIRS (REGISTRATION POOL, SEC. VII-B). THE REGISTRATION POOL (301 TEST PAIRS) DIFFERS FROM THE SCORED-PAIR SUBSET (255 PAIRS WITH VALIDITY-MASK CELLS). COUNTS REFER TO THE FROZEN BENCHMARK SNAPSHOT; SEE SEC. III-B FOR CURRENT-LABEL DRIFT.

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301

For assignment F_1 , the Hungarian algorithm maximizes total score within each query-image/gallery-image block; assignments below the validation-chosen threshold are rejected. Correct assignments are true positives, incorrect accepted assignments are false positives, and unmatched known queries are false negatives. Thus $F_1 = 2TP/(2TP + FP + FN)$; unmatched unknown queries are correct rejections. Unscorable pairs count as predicted negative for pairwise and assignment F_1 and rank last for retrieval. A known query with no finite score is a miss, while an unknown query with no finite score is not a false alarm.

Every metric is reported against its chance level, because two are far from zero (prior pool values; TBD for $n_Q=305$): pairwise F_1 has a best-constant-predictor floor $2p/(1+p) = 0.305$ on the prior pool, and Rank-5 chance is 0.767; Rank-1 and DIR@FAR chance are 0.336 and 0.038 (closed form, plus 200 random matrices through the same evaluation code). We call $2p/(1+p)$ a floor, not coin-flip chance: it is the score of predicting every pair positive, and it changes when the evaluated pair pool changes. Confidence resampling uses walls rather than queries because queries from one wall share capture conditions. With 11 test walls, no numeric interval is attached; wall-level intervals await a larger dataset and release of the per-pair score matrices (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The ceiling motivates a change of scope. A crack is a mark at a fixed location on a rigid, near-planar surface that carries texture the crack does not; we identify a crack by *where it is on its wall* once the two photographs are aligned. We release this as a deliberately simple *reference baseline* for CRACKID, with no learned component beyond the shared crack segmenter. The segmenter is a UNet++ model (`segmentation-models-pytorch`, `batch_segment.py` / `tiled_inference.py`) whose binary masks are filtered into instances with minimum area 200px and morphological closing 5px; the same masks and

Same identity, frame 0001



Stippled render



Same identity, frame 0009



Smooth painted plaster



Fig. 1. Examples from real CRACKID captures. Top: the same annotated identity in non-adjacent frames of one walk-around. Bottom: the two surface families, stippled render and smooth painted plaster. The identity labels inherit the dataset’s homography-based propagation and are illustrative, not an independent annotation audit.

instance extraction serve every method, so recall failures affect all rows identically. All baselines are zero-shot on CRACKID: no backbone is fine-tuned on crack pairs.

A. Alignment driven by non-crack texture

A wall is planar, so the homography $x' \sim Hx$ mapping photograph I_a into I_b ’s frame is the correct model. The design decision that matters is what drives the fit: we dilate the crack masks by 15px and exclude those pixels from detection, so H comes from wall texture alone. This is not an optimization but what makes the projected position an independent identity cue: were cracks to participate, a long crack could dominate the correspondence set and pull the fit toward the wrong

crack. Detection runs on copies downscaled so the long side is at most 1600px, after CLAHE equalisation (clip limit 2.0, 8×8 tiles, OpenCV). We use SIFT with up to 8000 features (OpenCV), FLANN matching with a 0.75 Lowe ratio test, and USAC_MAGSAC where available (falling back to RANSAC) at 4px with up to 10 000 iterations, requiring 25 inliers.

B. A cascade for two surface families

One front end does not serve both surfaces, and the failure is categorical: On the smooth-plaster walls, crack-masked SIFT succeeds on *exactly zero* pairs. We try three stages in order: (i) SIFT on masked wall texture, for stippled render; (ii) SIFT with cracks restored, triggered only when masking

starves detection below 300 keypoints on the weaker side (on plaster the crack is the only feature in frame), accepting the circularity stage (i) avoids, and recording which front end produced each result; (iii) ECC photometric alignment over a $320 \rightarrow 640 \rightarrow 1280$ px pyramid (translation then Euclidean initialisation, up to 300 iterations per level), accepted only if enhanced correlation exceeds 0.80. Every candidate H is sanity-checked (reject mirrored or folded quadrilaterals, area scale outside $[1/16, 16]$, collapsed sides); its full-resolution inlier RMS sets the Chamfer tolerance for that pair (Sec. VII-C). Stage identities (`sift`, `sift-nocrackmask`, `ecc`) are logged per image pair.

C. Projection and Chamfer agreement

Crack instances are connected components; pixels from I_a are projected by H and compared with instances of I_b . Chamfer distance, not IoU, is the criterion: cracks are a few pixels wide, so a 3 px registration error drives IoU to zero for a correctly aligned pair. Let $d(p, b) = \min_{q \in b} \|p - q\|_2$ be the point-to-set Euclidean distance and $|a|$ the pixel count of mask a . The reported score is the **symmetric mean Chamfer distance** in full-resolution pixels

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

over the union bounding box, with cheap rejection when boxes are far apart. We also report a scale-free alternative, **Chamfer coverage**, the symmetric fraction of pixels within τ , $\frac{1}{2}(|\{p \in a : d(p, b) \leq \tau\}|/|a| + |\{q \in b : d(q, a) \leq \tau\}|/|b|)$, with τ set per pair from that pair's own inlier RMS, clamped to $[2, 16]$ px (default 6.0 px when RMS is unavailable; the synthetic HYBRID-SKEL variant fixes $\tau=8.0$ px). Eq. (1) is a per-pair pixel distance whose scale differs per pair, so a single global threshold makes ranking partly measure calibration (Sec. VII-C).

D. Explicit non-answers, and the licensed negative

Correspondence is a one-to-one assignment per image pair with an explicit rejection threshold, not a per-query argmax: two cracks must not both claim the same crack; unmatched instances are reported as *disappeared* or *newly appeared*. The Hungarian solve [35] is part of *evaluation*, applied identically to every method's score matrix. Specific to this formulation is the evidence a **licensed negative** can carry: with I_a projected into I_b 's frame, the system observes a region visible and empty of the recorded crack, positive evidence of absence, categorically stronger than the low similarity a crop embedding produces. A projected crack falling outside the gallery photograph is a confident rejection, counted separately from unscorable pairs; this is the mechanism behind the open-set gap (Sec. VII-B). Unscorable pairs (no registration, or no retained instance after the 200 px size filter) receive $-\infty$ and are charged as predicted negative for F_1 and rank last for retrieval.

VI. THE APPEARANCE CEILING

A. Setup

Ten crop-scope matchers run under an identical protocol, identical inputs, and identical scoring: handcrafted keypoints (SIFT [14], ORB [15]), learned correspondence (SuperGlue [16] indoor weights, LoFTR [17] indoor weights, both at 512 px long side), general embeddings (ViT-B/16 [18], DeiT-S [19], CLIP ViT-B/32 [20] `openai/clip-vit-base-patch32`, DINOv2 [21] ViT-S/14, YOLOv8 [24] `yolov8n.pt` layer 9), and a re-identification architecture (OSNet [3] `osnet_x1_0` default weights). All image backbones are zero-shot on CRACKID with ImageNet or web pretraining only; no backbone is fine-tuned on cracks. Embedding crops are letterboxed to 224×224 with ImageNet normalisation and scored by cosine similarity; keypoint and learned correspondence scores are RANSAC inlier counts under the same validity mask. LightGlue [22] and NetVLAD [23] are cited as natural stronger retrieval front-ends but are not benchmarked here; they remain future baselines given compute and scope limits.

B. Result: appearance buys almost nothing on the decision

On the decision a record actually makes, same crack or not, at a threshold, no new crop row clears the prior best-constant floor by more than 0.069 (excess over the prior 0.305 floor at the oracle threshold, Table II: new F_1^* 0.337 to 0.374, i.e. $+0.032$ to $+0.069$; floors TBD pending rerun on the $n_Q=305$ pool), spanning handcrafted keypoints to foundation-model embeddings and more than two orders of relative compute (0.006 s to 7.19 s per query in separate per-method runs, Table II; absolute timings are workstation-dependent). At a deployable (validation-calibrated) threshold, the nine new rows cluster at $F_1^{\text{@}v}=0.331\text{--}0.338$ (prior floor $+0.026$ to $+0.033$); no method holds the prior $+0.046$ LoFTR exception under the new pool, floors TBD. Ranking does not transfer to deciding: the nine new rows separate over a 0.243 range on Rank-1 (0.370–0.613), yet ViT takes best Rank-1 (0.613) while its oracle pairwise F_1^* sits only 0.039 above the prior floor, and SuperGlue (seventh on Rank-1) has the best new crop open-set rate (0.384). Ordering a gallery well and keeping a record are different capabilities. The collapse after validation calibration is consistent with threshold instability under a small, wall-clustered sample: methods can order candidates differently while their score distributions do not support a stable same-crack/not-same-crack boundary on held-out walls. This is an interpretation of the observed pattern, not a claim that calibration itself is the sole cause. With 19 matchable test identities and no wall-level intervals, no fine ranking among crop baselines is claimed.

One contributor is checkable from Table I alone. Validation pair prevalence is 0.292 (26 012/89 118) against 0.180 on test, so the F_1 -optimal threshold is not a property of a method but of the pool it was fitted on: the same scores that separate cracks on the denser validation pool place their boundary in the wrong place once prevalence falls, and every method shifts

TABLE II

MAIN RESULT, TEST SPLIT, $n_Q = 305$ QUERY INSTANCES OVER 11 WALLS. ALL METHODS SEE IDENTICAL INPUTS AND ARE SCORED BY IDENTICAL CODE. F_1^* IS ORACLE-THRESHOLDED ON TEST (CEILING, NOT DEPLOYABLE); ΔF_1^* IS ITS EXCESS OVER THE PRIOR 0.305 BEST-CONSTANT-PREDICTOR FLOOR ($2p/(1+p)$, NOT COIN-FLIP CHANCE; SEC. IV); $F_1^{@v}$ IS THRESHOLDED ON THE HELD-OUT VALIDATION WALLS AND IS THE DEPLOYABLE NUMBER. *scored* IS THE FRACTION OF VALID QUERY-GALLERY PAIRS RECEIVING A FINITE SCORE; UNSCORED PAIRS ARE CHARGED AS PREDICTED-NEGATIVE, AND THE QUERY IS STILL COUNTED (SEC. VII-B). COST IS RELATIVE WALL-CLOCK SECONDS PER QUERY (NEW CROP ROWS: TOTAL_s/305 FROM SEPARATE PER-METHOD RUNS; USE FOR ORDERING, NOT ABSOLUTE TIMING). CHANCE AND FLOOR LEVELS ARE FROM THE PRIOR $n_Q=280$ POOL AND ARE MARKED TBD PENDING A CHANCE_BASELINE.PY RERUN ON THE $n_Q=305$ POOL (SEC. IV). BOLD MARKS THE BEST CROP-SCOPE VALUE PER COLUMN ONLY (NEW ROWS ONLY); $\dagger$ CLIP-CROP, COVERAGE-ONLY, AND REG+CHAMFER ROWS ARE THE PRIOR $n_Q=280$ SNAPSHOT PENDING RERUN AND MUST NOT BE COMPARED NUMERICALLY TO THE NEW ROWS. THE COVERAGE-ONLY REFERENCE ROW ANSWERS YES TO EVERY REGISTERED PAIR WITHOUT EXAMINING A CRACK AND IS THE LEVEL THE GEOMETRY ROW MUST BE READ AGAINST. COUNTS REFER TO THE FROZEN BENCHMARK SNAPSHOT; CURRENT DATASET/LABELS/ HAS BEEN RE-EDITED SINCE (SEE SEC. III-B).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{@v}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope (new $n_Q=305$ rerun except CLIP)</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [24]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding †	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	0.351	0.582	0.48	27.92

by the same amount. The pairwise- F_1 floor moves with it, from 0.452 to 0.305. This is a mechanism, not a measurement: the threshold-free version of the claim (pairwise average precision, or AUROC against prevalence) would settle it, and we do not report it; releasing the saved score matrices it requires is listed as future work (Sec. VIII).

C. Is it the crack, or the wall behind it?

Two controls separate identity cue from field of view at fixed backbone: a *crop+context* condition (@ctx1: instance bounding box expanded by $1.0\times$ its size on each side) and a *crack-erased* condition (@ctx1@nocrack: the same widened view with the crack mask dilated by 5px and filled by Telea inpainting, radius 3px, so the silhouette is removed rather than blacked out). Context helps a lot: OSNet’s Rank-1 rises from 0.654 to 0.775, mAP to 0.595, open-set rate from 0.168 to 0.404, and its pairwise F_1^* from 0.309 to 0.354 (prior $n_Q=280$ snapshot; context controls pending rerun at $n_Q=305$); CLIP’s open-set rate rises from 0.143 to 0.311, but erasing the crack costs almost none of it: OSNet retains Rank-1 0.711, mAP 0.574 and open-set 0.393; CLIP retains Rank-1 0.600, mAP 0.502, open-set 0.286. Both erased views still outrank every tight-crop method: the retrieved cue is the wall, that is, location. Pairwise F_1 moves little with context: all four context variants lie between 0.346 and 0.361, at most 0.056 above the prior floor, over a change that gives OSNet +0.236 of open-set rate. We read the aggregate as a property of the data: methods with no shared inductive bias land near the floor, and wherever a crop method improves, the improvement

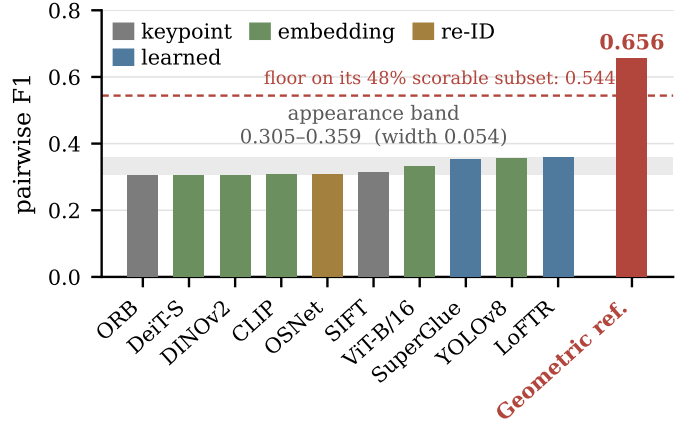


Fig. 2. The appearance ceiling. Pairwise F_1 for ten crop-scope matchers spanning four method families on CRACKID test ($n_Q=305$ for nine rows; CLIP-crop prior snapshot). New rows fall inside 0.337 to 0.374 against the prior 0.305 best-constant floor ($2p/(1+p)$) for the prior pool; floors TBD pending rerun, irrespective of family or relative compute. The geometric reference baseline (red, prior snapshot) leaves the band, but it answers only 48% of pairs, for which the best-constant floor is 0.544 (the COVERAGE-ONLY level), not 0.305: its bar is drawn against the dashed line and must not be read against the crop floor. Ranking metrics separate the same methods over a 0.24 range; the decision metric does not. All bars are point estimates on the rerun pool with no wall-level intervals; chance levels TBD.

is not the crack. Ten methods are a sample, not proof; the result establishes that the obvious moves buy 0.032 to 0.069 above the prior floor on the decision under the new rerun (floors TBD), reason to change the formulation, not the encoder.

VII. RESULTS

A. Main comparison and the open-set gap

On different pair pools, the geometric reference (prior $n_Q=280$ snapshot) changes the operating regime and is not numerically comparable to the new $n_Q=305$ crop rerun: prior crop Rank-1 0.654 (best) against geometric Rank-1 0.952, crop mAP 0.520 against 0.886, but the pools differ (*scored* 1.00 vs 0.48). Conditioned on coverage, as excess over the COVERAGE-ONLY reference, the prior ranking gains are $+0.437/+0.353$, while prior pairwise $F_1^* = 0.656$ is $+0.112$ over 0.544, not the $+0.351$ a comparison against the crop floor would suggest. The open-set result is the one number coverage does not explain: prior DIR@FAR=0.1 reaches 0.768 against 0.384 for the best new crop method (SuperGlue at $n_Q=305$), 0.404 for the best prior context control, prior chance 0.038 TBD, and exactly 0.000 for COVERAGE-ONLY. Observed: crop methods return a similarity to the nearest neighbor, which on self-similar hairline cracks is often high, so they cannot decline. Interpretation: an aligned pair supports the *licensed negative*: the query projects into the gallery frame, where its region is visible and empty. Relative compute does not explain the gap: LoFTR is the costliest new crop row (7.19 s/query), OSNet the cheapest (0.006 s/query) in separate per-method runs (Table II; absolute timings workstation-dependent).

B. Partial coverage, gap structure, and failures

Every crop method scores 100% of valid pairs, the geometric method 48%; Unscorable pairs are charged as predicted negatives. For ranking that is a penalty: censoring a complete matrix drops Rank-1 0.422 to 0.27 at 34% coverage in a held-out control (score matrices not committed; see paper README). For pairwise F_1 it is a subsidy: the scorable subset retains 100% of positives and 36.8% of negatives, so prevalence rises 0.180 to 0.373 and the floor with it, 0.305 to 0.544. COVERAGE-ONLY scores 0.544 pairwise F_1 and mAP 0.533, both above OSNet (0.309 and 0.520), and Rank-1 0.515, above ORB and SIFT but below the embedding and learned crop methods, by answering every registered pair without examining a crack. Its signal is registration overlap: it assigns one constant score to registered image pairs and $-\infty$ to unregistered pairs, so candidate images are ranked even though crack pixels are never inspected. Much of the ranking and pairwise- F_1 margin is therefore *which pairs align*; the open-set metric is the exception and the result we lead with.

Excluding the adjacent frame, then the two and three nearest, leaves Rank-1 flat (OSNet 0.654 to 0.648) with excess over chance stable in 0.24 to 0.32: the near-duplicate structure is real but the flat curve alone does not prove it is irrelevant, because the answerable pool shrinks (280 $\rightarrow$ 105) and prevalence shifts with it. It is also important for interpretation: the adjacent frame answers all 280 queries, whereas the gap-3 sweep leaves only 105 answerable queries, so the headline ranking measures short-baseline correspondence rather than broad temporal re-identification. Registration success is strongly wall-dependent (97/301 pairs, 32.2%: walls 07/08/12

100%, wall 09 52%, wall 11 83%, walls 13-16 between 3% and 29%, walls 10/17 at 0%). The pairs that register are the densely annotated ones, so the 96 registered pairs that carry a scored query-gallery pair hold 48% of the 31 822 valid ones — 160 valid cells per image pair against 104 for a failed pair — from a third of the available pairs (46 of the 301 same-wall pairs yield no valid pair at all). A logistic fit over all 301 pairs makes the mechanism plain: sharpness dominates (median 133.8 registered vs 7.2 failed, wall-clustered Wald $z = 5.1$, $p = 3 \times 10^{-7}$), frame gap far less ($z = -2.8$, $p = 0.005$). The independent-pairs z s (8.2, -5.0) overstate the contrast because failure is categorical by wall, so we resample walls, not queries, and the leave-one-wall-out spread in Rank-1 is large, so no fine ranking among baselines is claimed. One admission gate (variance-of-Laplacian ≥ 10) applied identically rejects 21 of 74 test captures and nearly doubles registration, 32.2% to 56.1%.

C. Ablations and cost

Chamfer distance versus coverage. The scale-free coverage score motivated in Sec. V is not a free improvement: it drops mAP 0.886 to 0.646 and Rank-1 0.952 to 0.849, the largest movement any change produces; DIR and $F_1^{@v}$ barely move (0.768 $\rightarrow$ 0.754, 0.582 $\rightarrow$ 0.556), so coverage gives a threshold decision, not a ranking. Masking. Disabling crack exclusion leaves ranking and pairwise agreement practically unchanged (to three decimals) but raises DIR@FAR 0.768 to 0.825: masking is not what buys the reported accuracy. This is expected when the test set contains no deliberate case where crack pixels create a spurious registration: masking buys evidence independence, not observed accuracy on this collection. We retain masking as the primary configuration because it keeps the alignment target disjoint from the identity target, and report the nomask $+0.057$ openly as a safeguard awaiting a harder stress test. ECC fallback. With ECC enabled (primary): Rank-1 0.952, mAP 0.886, *scored* 0.48; with ECC disabled: Rank-1 0.920, mAP 0.862, *scored* 0.50. Assignment F_1 rises 0.769 to 0.784 when ECC is enabled; open-set DIR is unchanged. ECC was kept versioned after a fix for a configuration in which it had been inert.

Cost. The geometric pass costs 7817 s total on test in our sessions (27.9 s/query if charged per query, about 26 s per same-wall image pair over 301 pairs). That per-query figure is a charging convention, not an intrinsic one: registration is per image *pair*, one homography serves every crack on a wall, and amortized over the 301 test image pairs the same pass runs about 26 s/pair, after which any later crack query on the pair pays only the Chamfer pass, not the homography. Absolute timings are workstation- and load-dependent and are reported for relative ordering only; relatively, the pair-level cost is within an order of magnitude of LoFTR, a crop method with none of the open-set benefit. No fine ranking by cost is claimed.

Protocol	Method	n_q	R@1	mAP	DIR@FAR	One threshold	Five thresholds
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
A (growth)	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.609
B (appearance)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
B (appearance)	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

TABLE III

HYBRID-SKEL VERSUS THE STRONGEST APPEARANCE BASELINE ON THE SYNTHETIC REVISIT PROTOCOL (SEC. VII-D); PAIRWISE AND ASSIGNMENT F_1 AT THE VALIDATION-CALIBRATED THRESHOLD. ONE SEED (0), TEN SOURCE PHOTOGRAPHS ACROSS FIVE WALLS, NO RESAMPLING; ROWS CARRY NO UNCERTAINTY AND ARE AN ACCURACY CHECKPOINT, NOT A STABILITY CLAIM (SEC. VII-D). HYBRID-SKEL IS NOT THE TABLE II REG+CHAMFER CONFIGURATION.

D. A synthetic revisit protocol

The operational setting a ticket management backend serves is a *revisited* crack — one that already has an open issue ticket — so we add a second protocol isolating controlled change. Ten source photographs across five of the walls (walls 01, 02, 04, 05, 09; a single seed, 0) are each synthesized into a set of queries under an identical affine warp ($1.08\times$ scale, 6° rotation, 4° tilt; affine approximation, not a full projective revisit): dataset A additionally grows a single tip of the largest crack component (length $0.18\times\max(16, W/14)$ px, width 3px, tip farthest from the component centroid, `hybrid_eval.py`), dataset B corrupts appearance while leaving structure untouched (Gaussian blur $\sigma=1.6$, noise $\sigma=14$, contrast/brightness shift); each synthetic query is ranked against the wall’s real instances only. Unlike the Table II REG+CHAMFER system, which is pure registration plus Chamfer scoring with per-pair RMS tolerance, the synthetic-protocol HYBRID-SKEL folds a structural skeleton comparison into the geometric baseline as a fallback plus per-pair blend ($\tau=8.0$ px), scored by the same code as OSNet+ctx (Table III). HYBRID-SKEL and REG+CHAMFER are therefore different systems; the synthetic rows test generalization under controlled change, not the Table II configuration. The warp is deliberately modest (real walk-around tilt reaches $p90\ 75^\circ$), so this is a checkpoint on small geometric change with controlled photometric or growth change, not a real revisit.

The geometric side does not transfer its main-benchmark advantage here: OSNet+ctx separates the synthetic queries at R@1 0.826/0.671, the HYBRID-SKEL at 0.275/0.274, with open-set rates 0.739/0.663 against 0.343/0.342. But the split is metric-dependent, and opposite the ranking gap: on identity agreement, the HYBRID-SKEL holds its own, with assignment $F_1^a = 0.646/0.640$ beating OSNet’s 0.631/0.609, and pairwise $F_1^{@v} = 0.569/0.574$ trailing 0.601/0.594 by far less than the ranking margin. Both rate 1.00 scored, and per assignment logs none was decided by the pure-homography branch alone ($H\% = 0.0$, the share of assignment cells whose verdict came straight from the homography with no skeleton fallback or blend, from the HYBRID-SKEL assignment logs rather than Table III): the geometric pipeline answers *which crack is which* on pairs appearance also resolves, but loses the global ranking of a hundred-plus same-wall candidates at

One threshold. These rows score a single-seed checkpoint on five walls, not a claim; they show the Table II advantage does not imply multi-visit robustness.

VIII. VALIDATION LIMITATIONS, AND FUTURE WORK

This benchmark establishes a first measurable formulation of crack re-identification; several directions remain open and represent natural extensions of the present work.

Scaling to multi-wall and multi-visit datasets. The test split covers 19 matchable identities across 11 walls, which is sufficient to establish the structural pattern (thresholded decisions near the floor; open-set benefit of explicit alignment) but not to rank individual methods with statistical confidence; all accuracy rows are point estimates with no wall-level intervals. A larger dataset — more walls, more surface families, and more annotators — would allow wall-level confidence intervals to be reported and method rankings to be interpreted reliably. More importantly, the current benchmark captures one continuous walk-around per wall; a second capture session at deliberately varied distances and lighting would convert the within-visit measurement into a genuine cross-visit benchmark. That real revisit data is the most valuable next step for assessing the re-identification component of operational ticket management backends.

Annotation quality and provenance-free labelling. Annotation quality was actively managed: a single human annotator placed and reviewed all 2,095 clicks from homography-propagated prefill until zero provisional points remained, and every one of the 1,023 clicks no component claimed was subsequently triaged against the imagery — 916 (89.5%) sit on visible cracks the segmenter missed rather than labelling mistakes, leaving 107 clicks (10.5%, 95% CI [8.7%, 12.5%]) naming no crack. Single-annotator; agreement unmeasured. The reported geometric advantage must therefore be read as provenance-conditioned until a small, manually verified subset labeled entirely without homography propagation quantifies how much is independent of annotation provenance. Releasing the per-pair score matrices alongside the dataset would additionally allow third parties to run sensitivity analyses on how much false-positive mass would be needed to manufacture an open-set gap of the observed magnitude, and to compute wall-clustered intervals and threshold-free AP/AUROC.

Closing the registration coverage gap. The geometric baseline currently answers 48% of query pairs and scores zero on smooth-plaster walls. Future work should explore surface-adaptive registration strategies — including learned local features, direct alignment on photometric gradients, or sparse depth from consumer devices — that could push coverage toward full-set evaluation. A learned front-end replacing SIFT would also decouple the geometry pipeline from the homography-based annotation provenance.

Learning to re-identify from geometry-derived supervision. The geometric baseline is a deliberate non-learned reference; it is not a deployment target. A natural next step is to use the registration pipeline as a teacher to generate pseudo-labels for

training a learned matcher, or to fine-tune a foundation-model backbone on crack pairs with homography- derived correspondence. This would let the system inherit the location-based advantage while remaining applicable to pairs the registration pipeline cannot answer.

Integration into ticket lifecycle evaluation. End-to-end evaluation in a ticket management context requires measuring not just retrieval accuracy but downstream impact: how often does a mis-identification open a duplicate ticket, and how often does a false match suppress a new defect? Future work should define task-level metrics (duplicate rate, missed-new-defect rate) and evaluate the full pipeline — segmentation, re-identification, and ticket assignment — in a closed-loop simulation that reflects real inspection workflows.

IX. CONCLUSION

We built a benchmark that makes crack re-identification testable: CRACKID, 140 photographs over 17 walls with 212 identities in the frozen snapshot, its annotation provenance (45% merged) stated rather than assumed, with the triage of unclaimed clicks complete (107/1023 naming no crack). All accuracy rows are point estimates on 19 matchable test identities with no wall-level intervals; we claim the pattern, not the digits. The benchmark’s main finding is negative and structural: on the decision a record requires, ten zero-shot matchers across four families buy between $+0.032$ and $+0.069$ over the prior best-constant floor under the new $n_Q=305$ rerun (new F_1^* 0.337–0.374; floors TBD), with no single-method exception at the validation threshold ($F_1^{@v}$ 0.331–0.338). Ordering a gallery and deciding an identity are not the same capability. The difficulty is not crack appearance alone: a geometric reference baseline (prior snapshot) that registers on wall texture with cracks masked out and Chamfer-matches projected masks reaches $\text{DIR@FAR}=0.1$ of 0.768 against 0.384 for the best new crop method and 0.404 for the best prior context method while answering 48% of pairs, and a COVERAGE-ONLY control exploiting that alignment without examining a crack scores 0.000 on the open-set metric (though 0.544 on pairwise F_1 by prevalence shift). Operationally, the open-set advantage comes from an aligned pair supporting a *licensed negative*, positive evidence of absence, where a crop embedding can only report low similarity; whether that advantage is independent of homography-based annotation remains unresolved and must be read as provenance-conditioned. Two bounds remain addressable: a real second capture session turns the benchmark from viewpoint to cross-visit, and a second annotator plus a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48% coverage is also a deployment gap in its own right, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the small single-seed synthetic revisit checkpoint reverses the headline ranking and open-set result in favour of OSNet+context, while the distinct HYBRID-SKEL pipeline retains only a narrow assignment- F_1 advantage; the present paper therefore establishes a promising

within-visit geometric reference, not a solved re-identification component for a multi-visit ticket management backend.

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